Model Family Is Recoverable From Co-Failure Alone: Population-Scale Provenance Signal Without Training-Data Access

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Abstract

Contributive and corroborative attribution answer "which training data explains this output," one output at a time, and need access to training data or model internals that is unavailable for most open leaderboard submissions. We ask whether anything about provenance survives in the only signal such a population does expose: which items each model gets wrong. Conditioning a model-by-item failure matrix on the exact, fit-free null implied by Rasch sufficiency and taking the leading eigenvectors of the residual correlation matrix, a 5-nearest-neighbour classifier recovers a model's base family with leave-one-out accuracy 0.807 over 916 labelled models from five families, against a label-permutation null of 0.287 ± 0.017 and a majority-class rate of 0.382. The same pipeline run on a draw from the exact null returns 0.318, at chance, so the construction does not manufacture the signal. It is not an accuracy artifact: projecting model accuracy out of the loadings leaves accuracy at 0.809. It is partly, but not mostly, a near-duplicate artifact: deduplicating at pairwise agreement 0.95 leaves 0.787, and the much more aggressive 0.90 threshold leaves 0.551, still far above chance. We report one negative result prominently: margin conditioning is not what creates the signal, since unconditioned correlation eigenvectors already reach 0.760. We did not set out to build an attribution method, and this is not one; it is evidence that population-scale behavioural correlation carries recoverable provenance information, obtained on a population where a pre-registered lineage analysis had to be abandoned for lack of declared metadata.

1 The gap we are pointing at

Contributive attribution estimates the causal effect of specific training examples on a specific output (Koh & Liang, 2017; Park et al., 2023); corroborative attribution finds sources that support a specific output (Worledge et al., 2024). Both operate on one model and one output, and both need access, to training data or to internals, that an open leaderboard population does not provide. Thousands of models are submitted with undisclosed lineage. The question there is not "which example explains this generation" but "which models share unstated ancestry, and can that be seen at all from behaviour."

We met this question from the measurement side, while characterising correlated failure between open language models, and we met it as a dead end: a pre-registered lineage decomposition had to be abandoned because the metadata it needed does not exist at usable coverage (Section 4). This paper reports what we did instead, and what it found.

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Arm	What it tests	Acc.	Perm.	Reading
Main	residual eigenvectors, real labels	**0.807**	0.287	signal present
Exact-null draw	pipeline on a curveball replicate	0.318	0.289	at chance
Accuracy only	accuracy as the sole feature	0.432	0.288	carries some, not most
Accuracy proj. out	loadings orthogonalised to accuracy	0.809	0.286	not an accuracy artifact
Raw correlation	unconditioned eigenvectors	0.760	0.287	conditioning not the source
Dedup 0.95	$N = 2{,}350$, 591 labelled	0.787	0.276	survives
Dedup 0.90	$N = 1{,}288$, 301 labelled	0.551	0.264	weakened, above chance

Table 1: Pre-registered controls. The exact-null arm (KW3) is the calibration check: because the curveball replicate destroys any real association between behaviour and family while preserving both margins, a pipeline that reported signal there would be reporting an artifact. It does not.

2 Setup: the exact conditional null and its residual spectrum

Let $F \in \{0,1\}^{N\times M}$ with $F_{im} = 1$ iff model i fails item m, row sums r_i, column sums c_m.

Lemma 1 (Mean co-failure depends only on item margins)**.** $O = \frac{1}{MN(N-1)} \sum_m (c_m^2 - c_m)$*, so any randomization preserving the item margins leaves the mean pairwise co-failure rate exactly invariant, with zero variance.*

Proof. $\sum_{i\neq j} F_{im}F_{jm} = (\sum_i F_{im})^2 - \sum_i F_{im}^2 = c_m^2 - c_m$ for $F_{im} \in \{0,1\}$. □

Lemma 1 says the first thing one would compute carries no provenance information at all once item difficulty is accounted for; the same degeneracy is known in ecology and psychometrics (Schluter, 1984; Gotelli, 2000; Cronbach, 1951). We therefore work with second-order structure. The row and column margins of F are the jointly sufficient statistics of the Rasch model, so the conditional law of F given its margins is uniform on the margin-matched set for every parameter value (Rasch, 1960; Andersen, 1973); we sample it with curveball (Strona et al., 2014; Carstens, 2015). Writing P for the Rasch mean field matching the margins, $D = (FF^\top - PP^\top)/M$ and R for D scaled to unit diagonal, we take the top-k eigenvectors of R, with k the number of eigenvalues exceeding the exact null's spectral edge.

Population. The current harvest of the archived Open LLM Leaderboard, ARC-Challenge: $N =$ 3,762 models, $M = 1{,}169$ items, read from result files at zero inference cost. On this population $k = 4$ (null edge 79.5) and the residual correlation is $5.4\times$ its null. This is a larger, later harvest than the $N = 1{,}362$ sample in our companion manuscript, and every number here refers to $N = 3{,}762$.

Labels. Base family is assigned by string matching on the model identifier against a list of known base-model names. This is a coarse lower bound on relatedness and explicitly not lineage ground truth; it attributes 34.5–35.7% of identifiers per benchmark. We use the five largest families, 916 labelled models: Mistral 350, Llama-2 211, Llama-3 188, Qwen 84, Mixtral 83.

3 Result: family is recoverable from co-failure

Figure 1 shows the effect directly. Leave-one-out 5-nearest-neighbour classification in the standardised k-dimensional loading space recovers family at accuracy $\mathbf{0.807}$, against a label-permutation null of 0.287 ± 0.017 (1,000 permutations, $p = 0.001$, the resolution floor) and a majority-class rate of 0.382. Recall is above chance for four of five families: Mistral 0.926, Llama-2 0.839, Llama-3 0.777, Qwen 0.714, Mixtral 0.386. Accuracy rises monotonically with the number of eigenvectors used, from 0.452 at $k = 1$ to 0.864 at $k = 6$ and 0.905 at $k = 20$, so the signal is distributed across many residual dimensions rather than carried by one.

Every kill condition and control was specified before the experiment ran.

What we are careful not to claim. Three things in Table 1 cut against the strongest reading, and we state them rather than bury them. First, margin conditioning is *not* what produces the signal: unconditioned correlation eigenvectors already reach 0.760, so conditioning adds about five points,

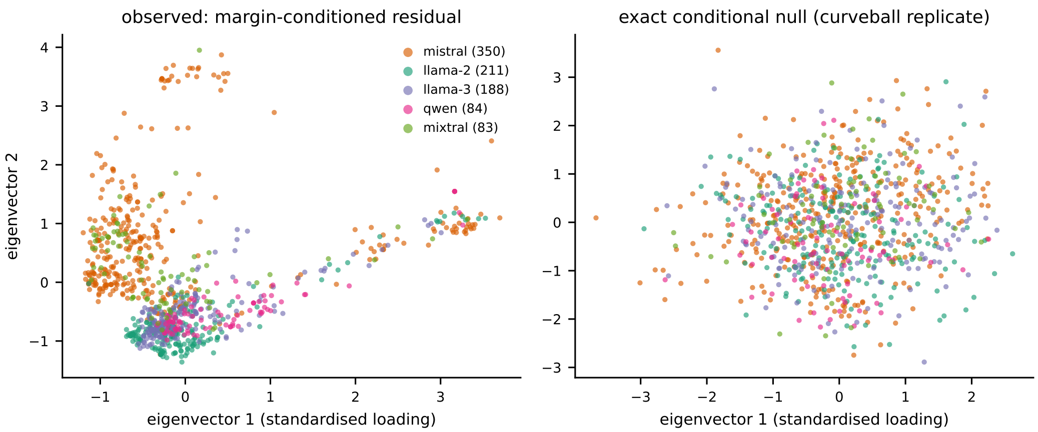

Figure 1: Models projected on the top two eigenvectors of the margin-conditioned residual correlation matrix, coloured by base family (left), and the identical pipeline applied to a curveball replicate of the same matrix (right). The replicate preserves both margins exactly and so destroys any real association between behaviour and family: it is what the left panel would look like if the construction were manufacturing the structure. Only the five largest families are drawn; the classifier of Section 3 uses four eigenvectors, not the two shown here.

not the effect. The exact null earns its place here as the calibration arm and as what makes the mean-level statistic provably uninformative (Lemma 1), not as the source of the result. Second, part of the signal is near-duplicate structure: accuracy falls from 0.807 to 0.551 under aggressive deduplication at agreement 0.90, which discards two thirds of the population. The residue is still roughly twice the permutation null, so the signal is not only duplicates, but it is partly duplicates. Third, the labels are name strings. A model named for a base it was not derived from, or derived from a base it is not named for, is mislabelled here, and we have no way to audit that.

4 The provenance opacity this was a response to

We did not choose an indirect route for its own sake. The direct one failed, and quantifiably. The leaderboard's optional, self-reported `base_model` field resolved a declared parent for 317 of $1{,}362$ models (23.3%) in our earlier harvest, with 315 repositories no longer retrievable at all. A 40% coverage threshold had been registered in advance as the point below which a lineage decomposition would not be trusted; it fired, and the decomposition was dropped rather than run on the surviving quarter. Name-string matching, used above, reaches 34.5–35.7% and is coarser still.

That failure is the reason this result matters for this workshop rather than only for the correlated-failure literature. On a population where declared provenance is missing for three quarters of models, a behavioural signal that recovers family at 0.81 from outcome data alone is the only provenance evidence available. It costs one Rasch fit and an eigendecomposition, needs no access to weights, training data, or inference, and scales to the full population.

5 What this is and is not

It is not an attribution method. It returns a coordinate in a residual eigenspace, not a claim about which examples caused which behaviour, and it has no notion of a specific output to explain. The natural next questions are whether the loadings support pairwise ancestry tests rather than family classification, whether the recovered structure survives against config-hash ground truth instead of name strings, and whether it degrades gracefully as families proliferate. What the result does establish is the premise those questions need: population-scale co-failure structure is not provenance-blind. Under an exact null that removes item difficulty by construction, and after projecting out model accuracy, behaviour alone still locates a model in its family most of the time.

6 Limitations

Single benchmark, single archive, one coarse labelling scheme; the five families used are the five largest and the sixth-largest onward are excluded. Deduplication at 0.90 costs the signal a third of its margin over chance, so a population with a different duplicate structure could behave differently. We report classification accuracy, not a calibrated provenance probability, and offer no decision procedure. An alternative explanation we cannot exclude is that the residual dimensions track training recipe or data mixture, which correlates with family without ancestry: that would make this a signature of convergent practice rather than of lineage, and distinguishing the two needs the ground truth this population lacks. Code, data, the pre-registration with its dated kill conditions, and every artifact behind the numbers above are archived and will be linked in the camera-ready. A large language model was used as a coding and drafting assistant; the authors specified and pre-registered the hypotheses and kill conditions, verified every derivation and every number against executed-run artifacts, and are responsible for the content.

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